Experiment 3: Forward Chaining Inference Engine (Data-Driven Reasoning)

Aim

To implement a forward chaining inference engine and use it to identify an animal from observed features, in the manner of the classic ZOOKEEPER system.

Theory

Forward chaining starts from the available data and applies rules to derive new facts until either a goal is reached or no rule can fire. It is data-driven: reasoning is triggered by what is known, not by what is asked. This suits monitoring, configuration and interpretation tasks where many conclusions may follow from the same data.

Each cycle re-computes the applicable rules against the growing fact base and fires one of them, adding its conclusion to the facts. The chain of fired rules (here R1 → R4 → R6) is itself valuable: it is the raw material for the explanation facility built in Experiment 7. Intermediate conclusions such as 'is mammal' and 'is carnivore' show how rule chaining builds abstraction level by level.

Algorithm

1. Initialise the fact set with the observed features.
2. Repeat: find all rules whose conditions are satisfied and whose conclusion is new.
3. If none exists, stop; otherwise fire the first such rule, add its conclusion, and record its identifier.
4. On termination, report the final fact set, the firing sequence, and the identification.

Procedure

1. Define the animal-identification knowledge base using IF–THEN production rules.
2. Initialise the working memory (WM) with the given observed features.
3. Match the conditions of each rule with the facts available in WM.
4. Construct the conflict set containing all applicable rules whose conclusions are not already present in WM.
5. Select the first rule from the conflict set according to the given rule order.
6. Fire the selected rule and add its conclusion to WM.
7. Record the identifier of the fired rule.
8. Repeat the matching and rule-firing process with the updated WM.
9. Continue until the required animal is identified or the conflict set becomes empty.
10. Display the final working memory, firing sequence, and identified animal.

Observations

The forward chaining inference engine starts with the observed features in working memory and progressively derives new facts by applying applicable IF–THEN production rules. After each rule is fired, its conclusion is added to working memory, which may make additional rules applicable in the next cycle.

The conflict set changes as new facts are derived. The recorded firing sequence shows how the inference engine moves from the initial observations to intermediate conclusions and finally to the specific animal identification. If no applicable rule remains before an animal is identified, the system reports that no identification can be established from the available facts.

Exercises

1. Add rules and facts to identify a tiger, a giraffe (long neck, long legs) and a zebra (black stripes, not carnivore).
2. Give the engine facts that satisfy no identification rule and make it report the most specific class it could establish.
3. Modify the engine to fire ALL applicable rules per cycle instead of the first one; compare the firing sequences.